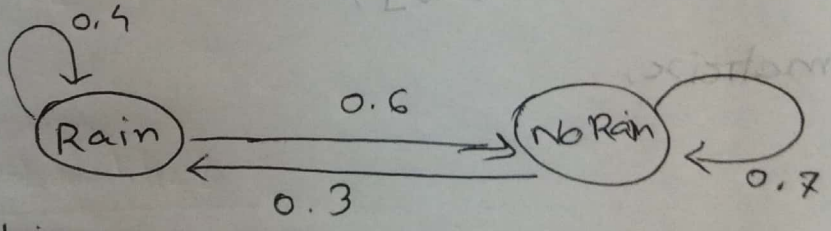
~~Type 1~~:

In probability theory, a Markov model is a stochastic model used to model randomly changing system over time.

Stochastic process is the study which involves ~~over~~ ~~time~~ how random variable changes over time.

$X_n = i \rightarrow$ means process is said to be in state i at time n.



transition matrix

$$\begin{array}{c} \\ R \\ N \end{array} \begin{array}{c} \begin{array}{cc} R & N \end{array} \\ \begin{bmatrix} 0.4 & 0.6 \\ 0.3 & 0.7 \end{bmatrix} \end{array}$$

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Markov chain is

It is the process of $X_0, X_1, X_2 \cdots$

- here system state are observeable

the ~~process~~ state of markov chain at time t is value of X_t.

for example

$X_t = 5$

the process is in state 5 at time t

Markov property →

$$P(X_{t+1}=j \mid X_t = i, X_{t-1} = i_1 \cdots X_0 = i_0)$$

$$= P(X_{t+1}=j \mid X_t = i)$$

for all $t = 1, 2, 3 \cdots$ and all state

Markov chain

Let $(X_0, X_1, X_2 \cdots)$ be a sequence of random variable, then $\{X_n, n \geq 0\}$ is a Markov chain if it satisfies the Markov

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property

$$P(X_{t+1}=j \mid X_t=i, X_{t-1}=i_1 \dots X_0=i_0)$$

$$= P(X_{t+1}=j \mid X_t=i)$$

(*) In a markov chain, the future depends only upon the present, Not upon the past.

Transition Probabilities

Probability from state i to state j after 1 step time period, denoted by P_{ij},

$$P_{ij} = P\{X_{t+1}=j \mid X_t=i\}$$

Ex → $P\{X_2=3 \mid X_1=2\} = P_{23}$

n – step probabilities

Probabilities from state i to state j after n-step time period, denoted by $P_{ij}^{(n)}$,

$$P_{ij}^{(n)} = P\{X_{n+1}=j \mid X_1=i\}$$

Ex → $P(X_3=3 \mid X_1=2) = P_{23}^{(2)}$

3-1=2

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Transition Probability matrix

The matrix describing Markov chain is called transition matrix.

$$P = \begin{array}{c} \\ 1 \\ 2 \\ \vdots \\ n \end{array} \begin{array}{c} \begin{array}{cccc} 1 & 2 & \cdots & n \end{array} \\ \begin{bmatrix} P_{11} & P_{12} & & P_{1n} \\ P_{21} & P_{22} & & P_{2n} \\ & & & \\ P_{n1} & P_{n2} & & P_{nn} \end{bmatrix} \end{array}$$

* It must be a square matrix
* $P_{ij} > 0$ for all i, j
* $\sum P_{ij} = 1$ (row wise)

$$\Rightarrow P_{n2} = P(X_2 = 2 \mid X_1 = n)$$

OR

$$P_{n2} = P(X_4 = 2 \mid X_3 = n)$$

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Notations

q_0 → Initial Probability of state

q_i → Probabilities of states after i time period

Get ~~markov~~

tmp of ~~mark~~ markov →

$X_n = i$

time ↙ ↘ state

$\Rightarrow P(X_n = a) = q_n(a)$

$\Rightarrow \boxed{q_n = q_0 P^n}$ *** or $q_{n+1} = q_n P$

$$P = \begin{array}{c} \\ a \\ b \\ c \end{array} \begin{array}{c} \begin{array}{ccc} a & b & c \end{array} \\ \begin{bmatrix} 0.1 & 0.5 & 0.4 \\ 0.6 & 0.2 & 0.2 \\ 0.3 & 0.4 & 0.3 \end{bmatrix} \end{array}$$

initial $(0.7, 0.2, 0.1)$

$q_0 = 0.7_a, 0.2_b, 0.1_c$

$P(X_2 = 3)$

time ↙ ↘ state

means ⟩ what is the probability of state 3 after time 2?

$P(X_2 = 3) = q_2(3)$

$q_2 = q_0 P^2$

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$$P^2 = P \cdot P$$

$$= \begin{bmatrix} 0.1 & 0.5 & 0.4 \\ 0.6 & 0.2 & 0.2 \\ 0.3 & 0.4 & 0.3 \end{bmatrix} \begin{bmatrix} 0.1 & 0.5 & 0.4 \\ 0.6 & 0.2 & 0.2 \\ 0.3 & 0.4 & 0.3 \end{bmatrix}$$

$$= \begin{bmatrix} 0.43 & 0.31 & 0.26 \\ 0.24 & 0.42 & 0.34 \\ 0.36 & 0.35 & 0.29 \end{bmatrix}$$

$$q_2 = q_0 P^2$$

$$= \begin{bmatrix} 0.7 & 0.2 & 0.1 \end{bmatrix} \begin{bmatrix} P^2 \end{bmatrix}$$

$$= \begin{bmatrix} 0.385 & 0.336 & 0.279 \end{bmatrix}$$

$$P(X_2 = 3) = 0.273$$

$$P(X_2 = 2 \mid X_0 = 1)$$

$$= P^2_{12}$$

$$= P \cdot P$$

$$= \begin{array}{c} \\ 1 \\ 2 \\ 3 \end{array} \begin{array}{c} \begin{array}{ccc} 1 & 2 & 3 \end{array} \\ \begin{bmatrix} 0.43 & 0.31 & 0.26 \\ 0.24 & 0.42 & 0.34 \\ 0.36 & 0.35 & 0.29 \end{bmatrix} \end{array}$$

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ii) $P\{X_3 = 2, X_2 = 3, X_1 = 3, X_0 = 2\}$

$2 \xrightarrow{1} 3 \xrightarrow{1} 3 \xrightarrow{1} 2$

$= q_0(2) * P_{23} * P_{33} \times P_{32}$

$= 0.2 \times 0.2 \times 0.3 \times 0.4$

$= 0.0048$

i) $P(X = 3)$

$= \sum_{i=1}^{3} P^{(1)}_{i3}\, q_i$

$= P_{13} q_1 + P_{23} q_2 + P_{32} q_3$

$= 04.$

i) $P\{x = 3\}$

$= \sum_{i=1}^{3} P_{i3}\, \pi_{0i}$

$= P_{13}\pi_{01} + P_{23}\pi_{02} + P_{33}\pi_{03}$

$= 0.4 \times (0.7) + 0.2 \times 0.2 + 0.3 \times 0.1$

$= 0.35.$

—o—

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18.3

$$P = \begin{matrix} 0 \\ 1 \end{matrix}\begin{bmatrix} 0 & 1 \\ 0.75 & 0.25 \\ 0.50 & 0.50 \end{bmatrix} \qquad m = 2$$

$$P^{(1)} = P \cdot P = \begin{bmatrix} 0.688 & 0.312 \\ 0.625 & 0.375 \end{bmatrix}$$

(a) $m = 1$ কারণ 0 থেকে শুরু তাইলে $0, 1 \to m$

$$\sum_{j=0}^{m=1} P_j = 1$$

$$P_0 + P_1 = 1 \quad \text{---(1)}$$

$$P_j = \sum_{i=0}^{m=2} P_i P_{ij}$$

$$P_0 = \sum_{i=0}^{m=1} P_i P_{i0}$$

$$= P_0 P_{00} + P_1 P_{10}$$

$$= P_0(0.75) + P_1(0.50)$$

$$\mu_{jj} = \frac{1}{P_j} \qquad j = 0 \cdots m$$

$$\mu_{00} = \frac{1}{P_0} = \frac{1}{0.75}$$

$$P_0 = P_0(0.75) + P_1(0.50)$$

$$1 - P_0 = P_1$$

$$P_0 = P_0(0.75) + 0.50 - P_0(0.50)$$

$$\Rightarrow P_0(1 - .75 + .50) = .50$$

$$\Rightarrow P_0 = \frac{.50}{.75} = \frac{50}{75} = \frac{2}{3}$$

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$$P_1 = 1 - \frac{2}{3} = \frac{1}{3}$$

$$\mu_{jj} = \frac{1}{P_j}$$

$$\mu_{00} = \frac{1}{P_0}$$

$$= \frac{3}{2} = 1.5$$

$$\mu_{11} = \frac{1}{P_1} = 3$$

—o—

18-4

$$P = \begin{array}{c} \\ 0 \\ 1 \\ 2 \end{array} \begin{array}{c} \begin{array}{ccc} 0 & 1 & 2 \end{array} \\ \begin{bmatrix} 0.6 & 0.2 & 0.2 \\ 0.3 & 0.4 & 0.3 \\ 0.0 & 0.3 & 0.7 \end{bmatrix} \end{array}$$

$$P_j = \sum_{i=0}^{m} P_i P_{ij}$$

$$P_0 = P_0 P_{00} + P_1 P_{10} + P_2 P_{20} = P_0(0.6) + P_1(0.3) + 0 P_2$$

$$P_1 = P_0 P_{01} + P_1 P_{11} + P_2 P_{21} = P_0(0.2) + P_1(.4) + P_2(.3)$$

$$P_2 = P_0 P_{02} + P_1 P_{12} + P_2 P_{22} = P_0(.2) + P_1(.3) + P_2(.7)$$

$$\sum_{i=0}^{m} P_i = 1$$

Again,

$$P_0 + P_1 + P_2 = 1$$

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$$P_0 = 0.6P_0 + 0.3P_1$$

$$0.3P_1 = 0.4P_0$$

$$P_1 = P_0 \frac{4}{3}$$

~~$1 = P_0 + P_1 + P_2$~~

$$P_1 = \frac{4}{3}P_0 = 0.2P_0 + 0.4 \times \frac{4}{3}P_0 + 0.3P_2$$

$$\Rightarrow .3P_2 = 0.6P_0$$

$$P_2 = \frac{6}{3}P_0 = 2P_0$$

$$P_0 + P_1 + P_2 = 1$$

$$\Rightarrow P_0 + \frac{4}{3}P_0 + 2P_0 = 1$$

$$\Rightarrow P_0 \left(1 + 2 + \frac{4}{3}\right) = 1$$

$$\Rightarrow P_0 \left(3 + \frac{4}{3}\right) = 1$$

$$\Rightarrow P_0 \left(\frac{13}{3}\right) = 1$$

$$P_0 = \frac{3}{13}$$

$$P_1 = \frac{3}{13} \times \frac{4}{3} = \frac{4}{13}$$

$$P_2 = \frac{6}{13}$$

$$\mu_{11} = \frac{1}{P_1} = \frac{13}{4}, \qquad \mu_{00} = \frac{1}{P_0} = \frac{13}{3}$$

$$\mu_{22} = \frac{1}{P_2} = \frac{13}{6}$$

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$$\mu_{ij} = 1 + \sum_{k \neq j} P_{ik} \mu_{kj}$$

$$\mu_{01} = 1 + P_{00}\mu_{01} + P_{02}\mu_{21}$$

$$= 1 + .6\mu_{01} + .2\mu_{21}$$

$$\mu_{21} = 1 + P_{20}\mu_{01} + P_{22}\mu_{21}$$

$$= 1 + 0\mu_{01} + 0.7\mu_{21}$$

$$\mu_{21} = 1 + .7\mu_{21}$$

$$.3\mu_{21} = 1$$

$$\mu_{21} = \frac{1}{.3}$$

$$= \frac{10}{3}$$

$$\mu_{01} = 1 + .6\mu_{01} + .2\left(\frac{10}{3}\right)$$

$$.4\mu_{01} = 1.67$$

$$\boxed{\mu_{01} = 4.167}$$

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